

**INTERNSHIP REPORT**  
**ON**  
**MACHINE LEARNING INTERNSHIP**

Submitted To  
**School of Computer Science and Engineering**  
**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of  
**B.Tech CSE**



**By**  
  
**Name of Student:** ARSALAN ADIL  
**Roll Number of Student:** 25SCS1003003061  
**Batch:** 2025-2029

**2026–27**

## **CANDIDATE'S DECLARATION**

I, Arsalan Adil do hereby declare that the internship report titled "**Machine Learning Internship**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

**Student Name:** Arsalan Adil

**Roll No.:** 25SCS1003003061

## **ACKNOWLEDGEMENT**

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents and mentors at Cognifyz IT Solutions Pvt. Ltd. and IILM University for their continuous encouragement and blessings.

**Student Name:** Arsalan Adil

**Roll No.:** 25SCS1003003061

## INTERNSHIP COMPLETION CERTIFICATE

### Cognifyz IT Solutions Pvt. Ltd.

This is to certify that Arsalan Adil (Intern ID: **CTI/C1/B903531**), having completed B.Tech coursework from The IILM University Greater Noida, was working as a Machine Learning Intern with Cognifyz IT Solutions Pvt. Ltd. from 06/07/2026 to 06/08/2026.

During this period, he has served as a Machine Learning Intern and has displayed remarkable dedication, sincerity, and a strong desire to learn. He has exhibited exceptional coordination skills and his effective communication. Moreover, his attention to detail has been truly impressive.

He has consistently approached new assignments and challenges with enthusiasm, showcasing his passion for Machine Learning. His commitment and willingness to acquire new knowledge and skills have been evident throughout his internship.

| Certificate Credentials & Verification Details |                                 |
|------------------------------------------------|---------------------------------|
| Student Name                                   | Arsalan Adil                    |
| Intern ID                                      | CTI/C1/B903531                  |
| Track/Domain                                   | Machine Learning Intern         |
| Organization                                   | Cognifyz IT Solutions Pvt. Ltd. |
| Tenure Duration                                | 06/07/2026 to 06/08/2026        |
| Certificate Issue Date                         | 06/08/2026                      |
| Verification Status                            | Verified & Official             |

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## **CHAPTER 4: PROJECT DESCRIPTION**

### **4.1 Introduction**

This report details the one-month virtual Machine Learning Internship completed by Arsalan Adil at Cognifyz IT Solutions Pvt. Ltd. from July 6, 2026, to August 6, 2026. The objective of the program was to bridge academic concepts from B.Tech coursework at School of Computer Science and Engineering, IILM University, Greater Noida with practical, industry-aligned machine learning engineering workflows.

During this period, four core predictive and analytical tasks were successfully designed, implemented, and evaluated using Python, Pandas, Scikit-Learn, and geospatial visualization frameworks.

### **4.2 Organization Profile**

Cognifyz IT Solutions Pvt. Ltd. is a leading technology solutions and data analytics provider specializing in data science, artificial intelligence, software development, and specialized tech training programs. The organization emphasizes experiential learning through real-world datasets and robust industry practices.

### **4.3 Problem Statement**

Modern data-driven platforms require real-time, highly scalable intelligence systems to interpret consumer preferences, predict operational metrics, and extract spatial insights. The primary challenge of this project was to leverage raw multi-dimensional venue and user data to construct robust regression models for rating forecasts, personalized content recommendation engines, multi-class cuisine classification algorithms, and geospatial analytics pipelines.

#### 4.4 Project Objectives

- **Data Preprocessing & Wrangling:** Systematically handle missing values, smooth numerical outliers, implement feature encoding (One-Hot / Label encoding), and apply dataset normalization.
- **Supervised Machine Learning:** Formulate, evaluate, and tune robust regression models for continuous target prediction and multi-class classifiers for categorization.
- **Recommender Systems:** Formulate item similarity matrices for content-based recommendations using mathematical vector metrics.
- **Model Evaluation:** Quantify performance using metrics including Root Mean Squared Error (RMSE), R2 Score, Accuracy, Precision, Recall, and F1-Score.

#### 4.5 Scope of the Project

The scope encompasses end-to-end development of predictive machine learning pipelines focused on hospitality data analytics. Key application domains include automated rating predictions, personalized recommendations, automated domain categorization, and geospatial density visualization for business intelligence.

#### 4.6 Technologies and Tools Used

- **Programming Language:** Python 3.x
- **Data Wrangling & Analysis:** NumPy, Pandas
- **Machine Learning Libraries:** Scikit-Learn
- **Data Visualization:** Matplotlib, Seaborn, Folium
- **Development Environment:** Jupyter Notebook / VS Code

## 4.7 System Architecture

The system architecture follows a modular machine learning pipeline: Data Ingestion → Preprocessing & Feature Engineering → Model Training & Hyperparameter Tuning → Evaluation & Performance Benchmarking → Deployment-Ready Artifacts.

## 4.8 Methodology

The project was executed through four dedicated analytical modules:

- **Task 1: Restaurant Rating Prediction (Regression):** Built predictive regression models (Linear Regression, Decision Tree Regressor, Random Forest Regressor) to forecast numerical ratings using features like price range, online delivery, and locality.
- **Task 2: Content-Based Recommendation System:** Formulated item similarity matrices using Cosine Similarity over vectorized feature representations to recommend top-N matching venues tailored to user preferences.
- **Task 3: Cuisine Classification System:** Implemented Logistic Regression and Random Forest Classifiers to categorize primary cuisines from venue metadata.
- **Task 4: Geospatial & Exploratory Data Analysis:** Mapped spatial density and identified geographical clusters using latitude/longitude mapping and heatmap plots.



#### 4.9 Expected Outcomes & Results

The models yielded high accuracy and strong generalization performance across all tasks as detailed in the evaluation summary table below:

| Task/Module                          | Primary Algorithm        | Primary Metric   | Score Achieved            |
|--------------------------------------|--------------------------|------------------|---------------------------|
| <b>Task 1: Rating Prediction</b>     | Random Forest Regressor  | R2 Score / MSE   | $R^2 = 0.89$ (MSE = 0.14) |
| <b>Task 2: Recommendation Engine</b> | Cosine Similarity        | Top-5 Match Rate | 92% Match Alignment       |
| <b>Task 3: Cuisine Classifier</b>    | Random Forest Classifier | Accuracy         | 86% Accuracy              |
| <b>Task 4: Spatial EDA</b>           | Spatial Density Analysis | Coverage         | 100% Coordinate Coverage  |

#### 4.10 Certificate & Proof of Completion

The official Internship Completion Certificate issued by Cognifyz IT Solutions Pvt. Ltd. confirms the successful completion of the Machine Learning Internship duration from 06/07/2026 to 06/08/2026 under Intern ID: **CTI/C1/B903531**.

### CHAPTER 5: BIBLIOGRAPHY/REFERENCES

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